

# Internship Final Report

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University: Central University of Jammu

Major: Computer Science

Internship Duration: June 1, 2026 - June 30, 2026

Company: ShadowFox

Domain: AI/ML

Mentor: Mr. Hariharan

Coordinator: Mr. Aakash

## Objectives

Develop and evaluate an image classification model using TensorFlow on the CIFAR-10 dataset.

## Tasks and Responsibilities

Prepared the dataset, designed a CNN model, trained it on Google Colab, evaluated its performance, and documented the results.

## Learning Outcomes

Learned image preprocessing, CNN architecture, TensorFlow model training, evaluation, and model saving.

## Challenges and Solutions

TensorFlow was not supported on Python 3.14, so Google Colab was used. Hyperparameters were adjusted to improve accuracy.

## Results

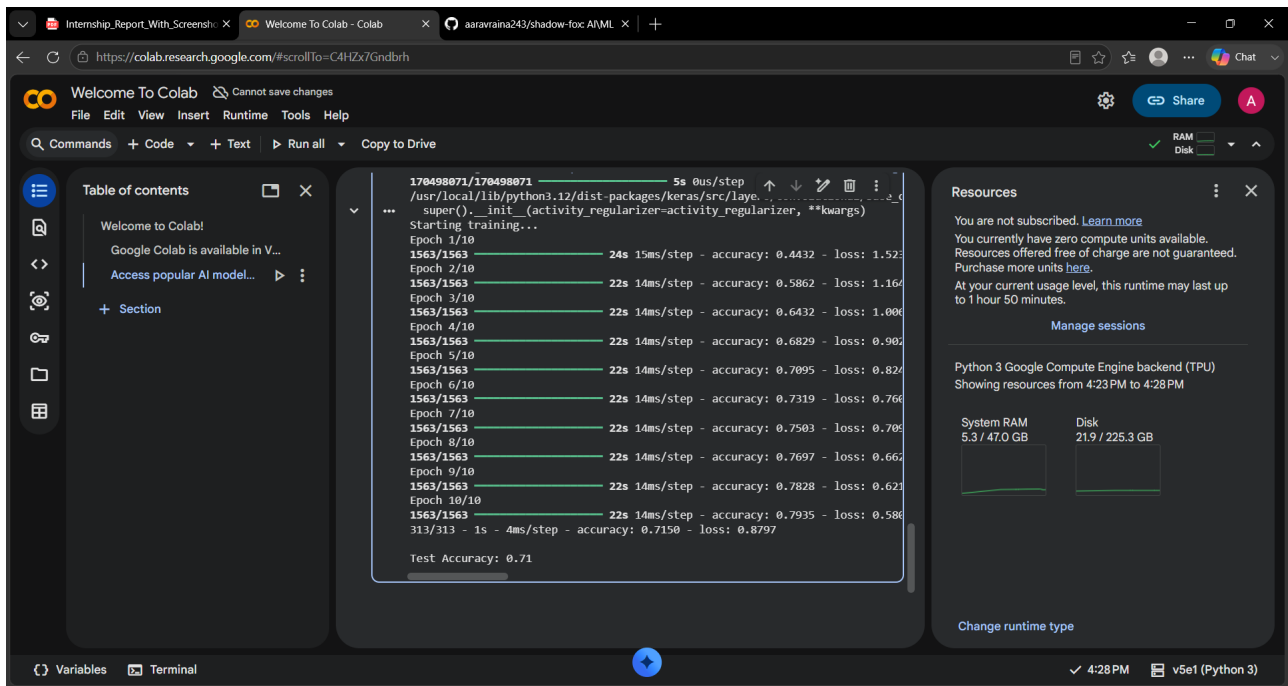


Figure 1: Model training progress and final test accuracy (~71%) on the CIFAR-10 dataset.

## Conclusion

The project successfully demonstrated image classification using a simple CNN with satisfactory accuracy for a beginner-level implementation.